

Mutual Fund Analytics Capstone Project

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1. Executive Summary

The Mutual Fund Analytics Capstone project was developed to analyze mutual fund industry trends, fund performance, investor behavior, and portfolio risk using Python, SQL, and Power BI. The project involved data ingestion, cleaning, exploratory analysis, performance analytics, dashboard development, and advanced risk analytics. A comprehensive Power BI dashboard was created to provide actionable insights into mutual fund investments and investor activity.

2. Project Objective

The primary objective of this project was to build an end-to-end Mutual Fund Analytics solution capable of:

- Analyzing mutual fund performance.
- Tracking industry growth and SIP trends.
- Understanding investor behavior.
- Evaluating risk-adjusted returns.
- Building an interactive business dashboard.
- Generating actionable investment recommendations.

3. Data Sources

The project utilized 10 datasets:

1. Fund Master Data
2. NAV History
3. AUM by Fund House
4. Monthly SIP Inflows
5. Category-wise Inflows
6. Industry Folio Count
7. Scheme Performance
8. Investor Transactions
9. Portfolio Holdings
10. Benchmark Indices

Additionally, live NAV data was fetched from the MFAPI public API for selected mutual fund schemes.

4. Technology Stack

Programming & Analytics

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly
- SciPy

Database

- SQLite
- SQLAlchemy

Visualization

- Power BI Desktop

Version Control

- Git
- GitHub

5. Architecture Diagram



6. ETL Process

Data Ingestion

All 10 CSV datasets were loaded using Pandas and validated for structure, completeness, and consistency.

Data Validation

The following validations were performed:

- Missing value analysis
- Data type validation
- AMFI code verification
- Duplicate record detection
- Date consistency checks

Data Cleaning

NAV Data

- Converted dates to datetime format
- Sorted by AMFI code and date
- Removed duplicate records
- Validated positive NAV values

Investor Transactions

- Standardized transaction types
- Validated transaction amounts
- Cleaned KYC status values
- Converted transaction dates

Scheme Performance

- Validated performance metrics
- Checked expense ratio ranges
- Verified Sharpe Ratio values

7. Database Design

A SQLite database was created containing:

- dim_fund
- fact_nav
- fact_transactions

- fact_performance
- fact_sip

The database enabled SQL-based analytical queries and dashboard integration.

8. Exploratory Data Analysis (EDA)

NAV Trend Analysis

Historical NAV data from 2022–2026 was analyzed for 40 mutual fund schemes.

Findings

- Strong recovery observed after market corrections.
- Equity funds demonstrated long-term upward trends.
- Small-cap funds exhibited higher volatility.

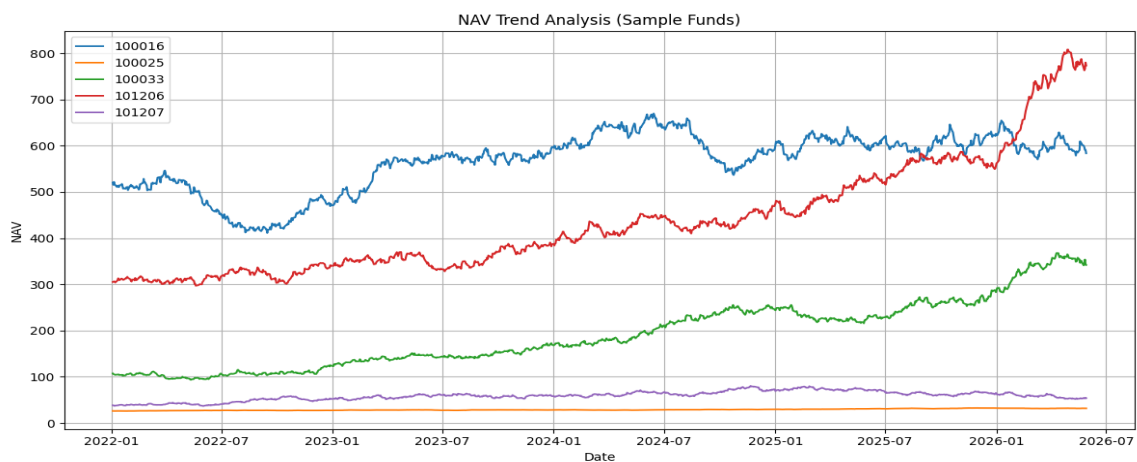


Figure 1: NAV Trend Analysis

AUM Analysis

Top fund houses by Assets Under Management:

- SBI Mutual Fund
- ICICI Prudential Mutual Fund
- HDFC Mutual Fund
- Nippon India Mutual Fund
- Kotak Mahindra Mutual Fund

SBI Mutual Fund maintained industry leadership in total AUM.

SIP Trend Analysis

Monthly SIP inflows increased significantly during the study period.

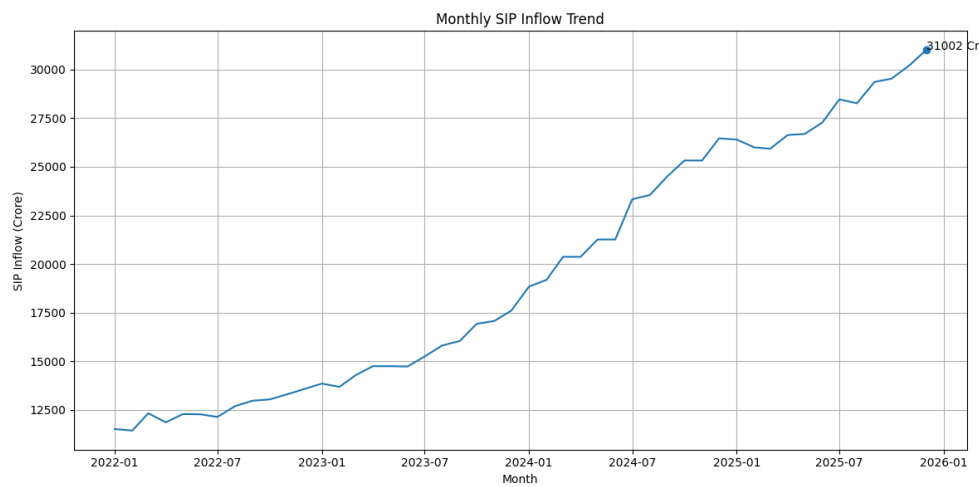


Figure 2: Monthly SIP Inflow Trend

Key Observation

Highest SIP inflow reached: ₹31,002 Crore

This indicates increasing retail investor participation.

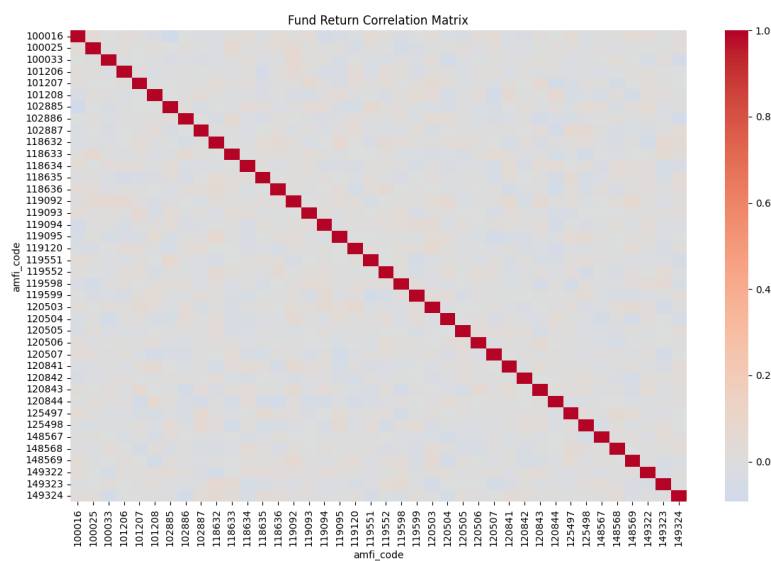


Figure 3: Correlation Matrix

Investor Analytics

Investor behavior was analyzed by:

- State
- Age Group

- City Tier
- Transaction Type

Key insights showed strong participation from T30 cities and investors aged 26–45 years.

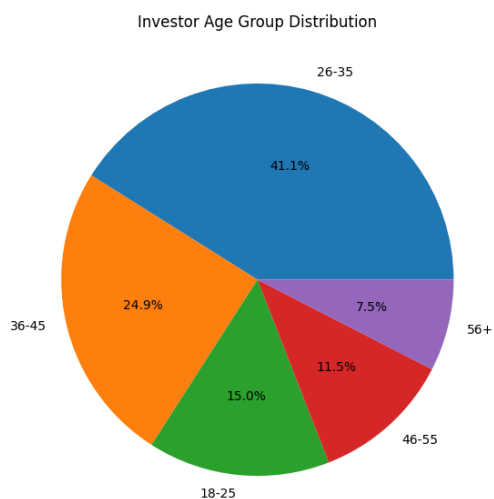


Figure 4: Investor Demographic Distribution by Age Group

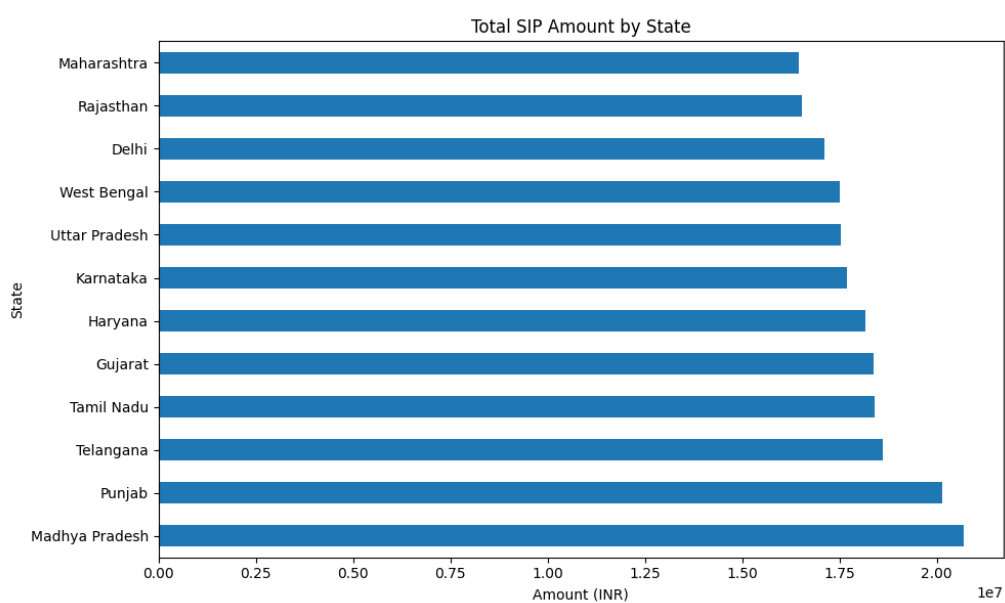


Figure 5: Geographic Distribution of Investments by State

9. Performance Analytics

Several advanced performance metrics were calculated.

CAGR

Compound Annual Growth Rate was calculated using NAV history.

High-growth equity funds delivered strong long-term returns.

Sharpe Ratio

Measured risk-adjusted performance.

Funds with higher Sharpe Ratios demonstrated superior risk-adjusted returns.

Highest Sharpe Ratio: 1.09

Sortino Ratio

Focused on downside risk.

Sortino Ratios identified funds capable of generating returns with lower downside volatility.

Highest Sortino Ratio: 1.83

Alpha & Beta

Alpha and Beta were computed against benchmark performance.

Findings:

- Positive Alpha funds outperformed benchmarks.
- High Beta funds demonstrated greater market sensitivity.

Maximum Drawdown

Maximum drawdown was calculated for all funds.

The metric helped identify downside risk during adverse market conditions.

Fund Scorecard

A composite score was developed using:

- Return Rank
- Sharpe Rank
- Alpha Rank
- Expense Ratio Rank
- Drawdown Rank

Best Fund Score: 100

10. Dashboard Development

A four-page interactive Power BI dashboard was developed.

Page 1: Industry Overview

Features:

- Total AUM KPI
- SIP Inflow KPI
- Industry Growth Trends
- AUM by Fund House

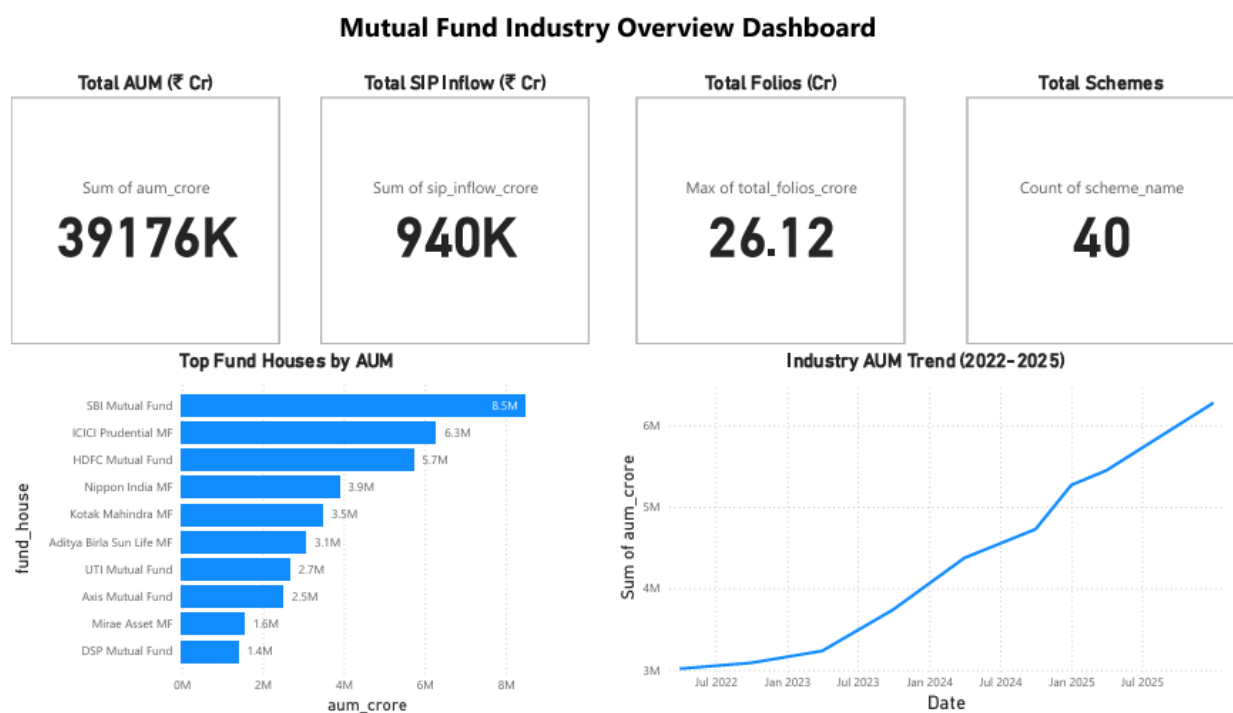


Figure 6: Industry Overview Dashboard

Page 2: Fund Performance

Features:

- Risk vs Return Scatter Plot
- Fund Scorecard
- NAV Trend Analysis
- Fund Selection Slicer

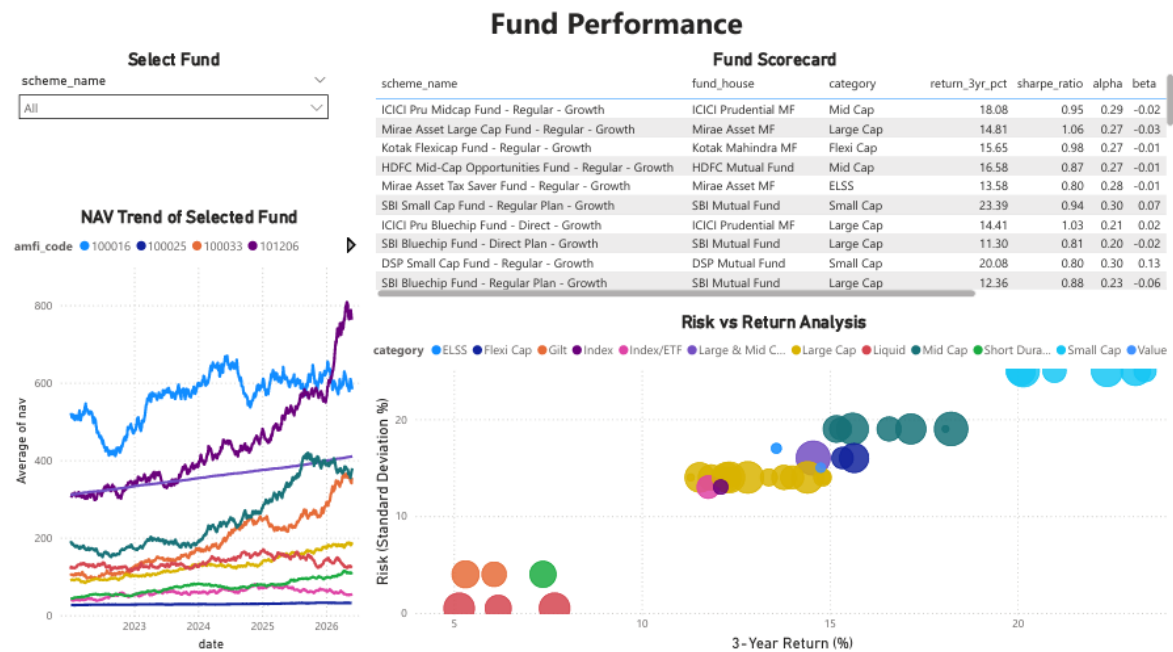


Figure 7: Fund Performance Dashboard

Page 3: Investor Analytics

Features:

- State-wise Investments
- Transaction Type Distribution
- Age Group Analysis
- Monthly Transaction Volume

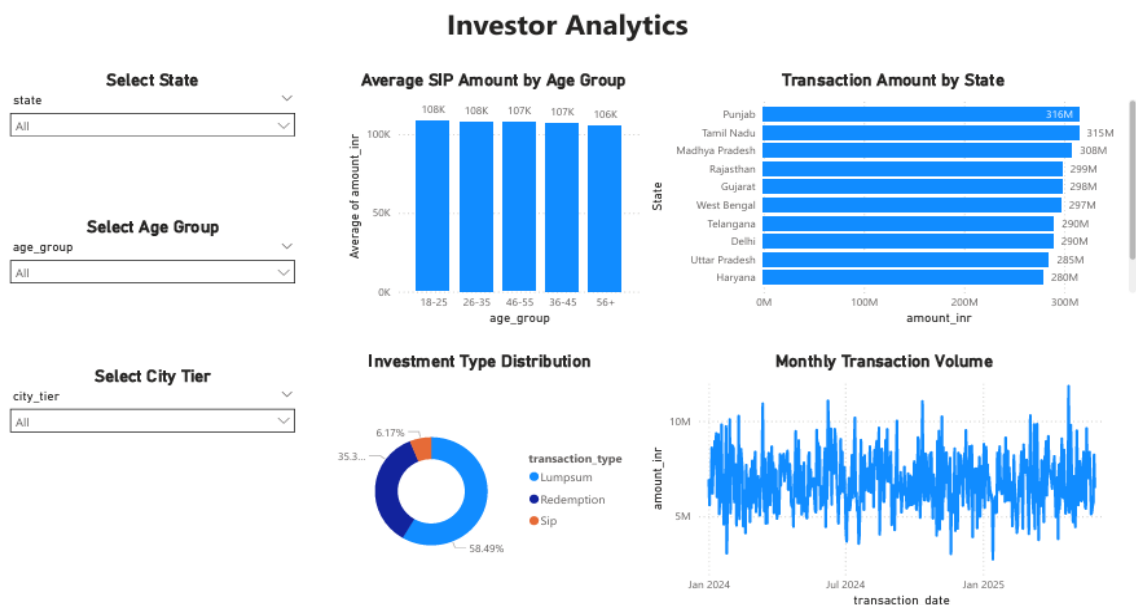


Figure 8: Investor Analytics Dashboard

Page 4: SIP & Market Trends

Features:

- SIP Inflow Trends
- Category Heatmap
- Top Categories by Net Inflow
- Industry KPIs

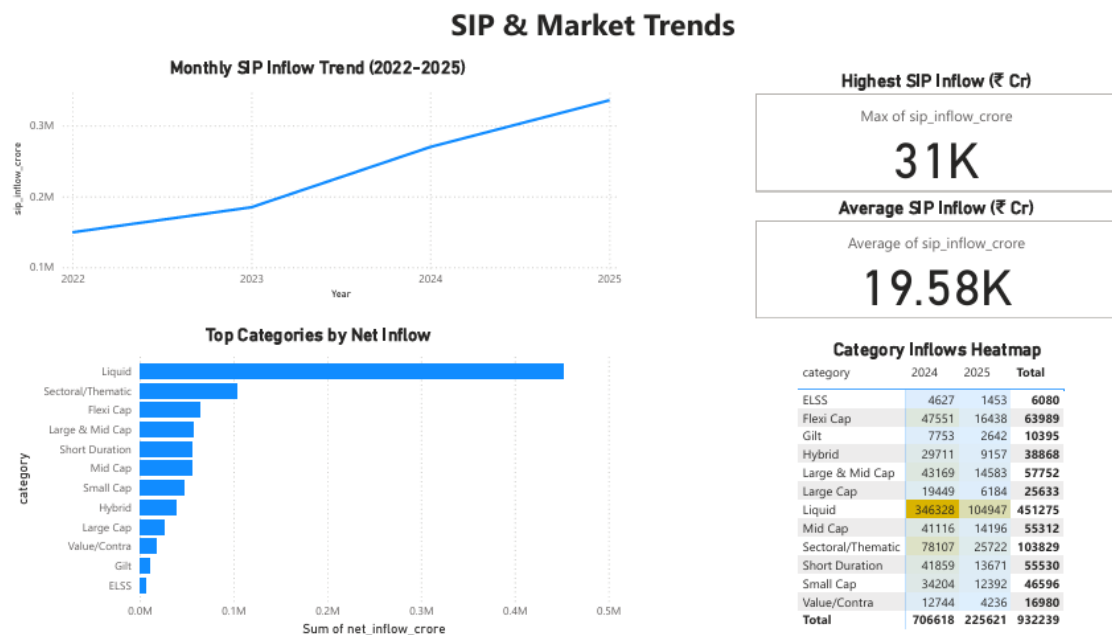


Figure 9: SIP & Market Trends Dashboard

11. Advanced Analytics

Historical VaR & CVaR

Calculated downside risk using historical returns.

Results highlighted differences in tail-risk across mutual funds.

Rolling Sharpe Ratio

Tracked risk-adjusted performance over time using rolling 90-day windows.

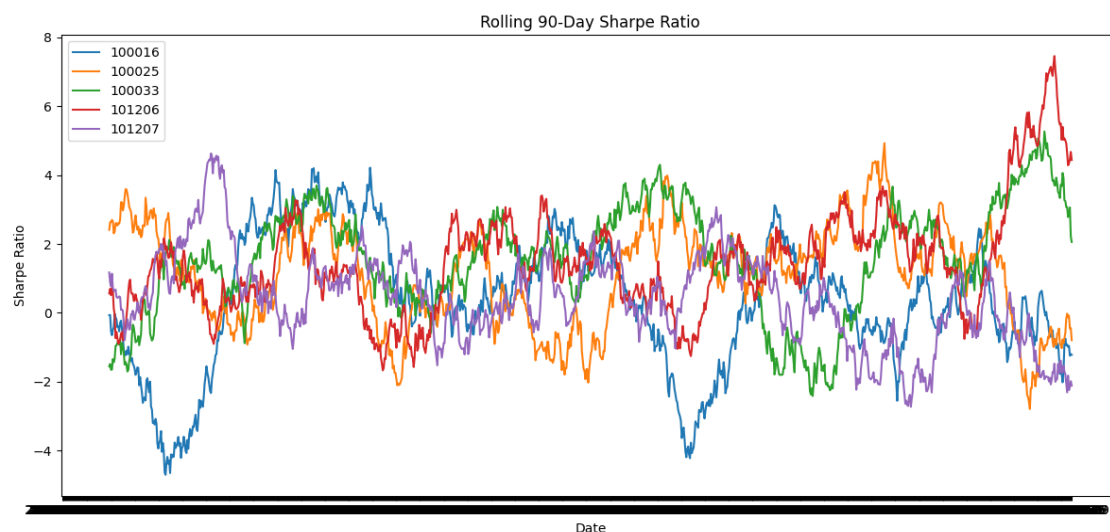


Figure 10: Rolling 90-Day Sharpe Ratio Trend

Investor Cohort Analysis

2024 cohort dominated both investment volume and investor participation.

SIP Continuity Analysis

Several investors exhibited average SIP gaps exceeding 35 days and were classified as At-Risk.

Fund Recommendation Engine

Developed a rule-based recommendation engine that suggests funds according to:

- Low Risk
- Moderate Risk
- High Risk

Sector Concentration Risk

Herfindahl-Hirschman Index (HHI) was calculated and values ranged from approximately 0.12 to 0.20, indicating moderate diversification across most portfolios

Results showed that most portfolios remained diversified while a few displayed moderate concentration risk.

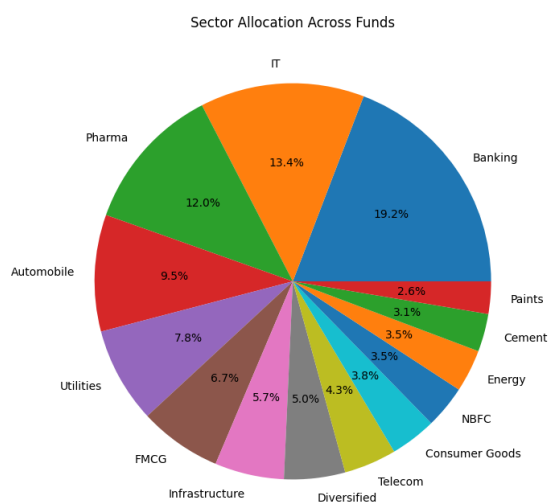


Figure 11: Sector Allocation Across Funds

12. Key Findings

1. Mutual fund industry AUM continued strong growth throughout the study period.
2. Retail SIP participation increased significantly.
3. SBI Mutual Fund maintained leadership in total AUM.
4. Several funds generated strong risk-adjusted returns based on Sharpe Ratio.
5. Small-cap funds delivered higher returns but with increased volatility.
6. Investor participation was concentrated among working-age demographics.
7. Portfolio diversification levels remained healthy across most funds.
8. Cohort analysis revealed strong investor retention among earlier investors.

13. Recommendations

1. Investors with low risk appetite should focus on liquid and debt-oriented schemes.
2. Moderate risk investors may prefer large-cap and blue-chip funds.
3. High risk investors may consider mid-cap and small-cap opportunities.
4. Fund managers should monitor SIP discontinuation risk.
5. Portfolio diversification should remain a priority to reduce concentration risk.

14. Limitations

- Historical data does not guarantee future performance.
- Public benchmark data may not perfectly reflect scheme benchmarks.
- Investor behavior was analyzed using available transaction records only.
- Some datasets contained synthetic values for academic purposes.

15. Conclusion

This project successfully implemented an end-to-end Mutual Fund Analytics platform covering ETL, SQL database design, exploratory analysis, performance analytics, advanced risk metrics, dashboard development, and investor analytics.

The final solution demonstrates practical applications of Python, SQL, Power BI, and financial analytics in solving real-world business problems and supporting investment decision-making.